

Configuring Remote Desktop on Ubuntu 24.04+

This guide provides two distinct methods for configuring remote desktop on Ubuntu 24.04+ using `gnome-remote-desktop`.

Understanding Remote Session Types

There are two main types of remote desktop sessions:

- **Mirrored Session (Screen Sharing):** This connects you to the session currently active on the physical machine's display. The resolution is locked to the physical monitor's resolution, and the remote connection terminates if the local user logs out. This is best for remote assistance.
- **Virtual Session (Headless):** This creates a brand new, independent desktop session for your remote connection. The resolution can be defined by your client, and you can connect even if no one is logged into the machine locally. This is ideal for server administration.

The key to switching between these modes is to use the correct `systemd` service: - The **system-wide service** (`gnome-remote-desktop.service`) runs with system privileges, is not tied to a user session, and allows for virtual sessions. - The **user-level service** (`gnome-remote-desktop.service` run with `--user`) is tied to the active user's login session and is used for mirroring.

This guide outlines how to configure each one.

Method 1: Headless Remote Login (Virtual Session)

Use this method when you want to connect to the server remotely, even if no user is physically logged in. It creates a new, virtual desktop session for you by using the system-wide remote desktop service. The display resolution can be set by your client (e.g., Remmina). This is the ideal method for servers.

Key Steps:

1. Ensure the user-level service is disabled.
2. Enable and configure the system-level service.
3. Enable user lingering to allow the service to run after reboots without a login.

Instructions:

1. Install GNOME Remote Desktop

If not already installed, this command installs the necessary software.

```
sudo apt-get update
sudo apt-get install -y gnome-remote-desktop
```

2. Disable the Conflicting User Service

This ensures the user-level service doesn't interfere with the system-wide one we intend to use.

```
systemctl --user disable --now gnome-remote-desktop.service
```

(It's okay if this command fails; it just means the service wasn't active for your user.)

3. Enable the System-Wide Service

This is the core step. It enables the service that runs with the `--system` flag, allowing for headless operation.

```
sudo systemctl enable --now gnome-remote-desktop.service
```

4. Configure the Firewall

Allow incoming RDP connections through the firewall.

```
sudo ufw allow 3389/tcp
```

5. Set RDP Credentials for the System Service

Use `sudo` and the `--system` flag to configure the system-wide daemon. Replace "your_password" with a strong password.

```
sudo grctl --system rdp set-credentials "your_username"
"your_password"
```

6. Enable Linging for Your User

This is critical. It allows your user's services to start at boot and run in the background, which is required by the remote desktop daemon.

```
sudo loginctl enable-linger your_username
```

7. Reboot

A reboot is the most reliable way to ensure all changes, especially lingering and the new system service, are properly activated.

```
sudo reboot
```

Method 2: Mirrored Remote Desktop (Screen Sharing)

Use this method when you want to connect to and control the desktop session that is currently active on the physical monitor. It works by enabling the user-level remote desktop service. The resolution is determined by the server's monitor. This is best for providing remote assistance to a user who is already logged in.

Key Steps:

1. Ensure the system-level service is disabled.
2. Enable and configure the user-level service.

Instructions:

1. Install GNOME Remote Desktop

If not already installed, this command installs the necessary software.

```
sudo apt-get update
sudo apt-get install -y gnome-remote-desktop
```

2. Disable the Conflicting System Service

This ensures the system-wide service doesn't interfere with the user-level one we intend to use.

```
sudo systemctl disable --now gnome-remote-desktop.service
```

(It's okay if this command fails; it just means the service wasn't active.)

3. Enable the User Service

This command is run without sudo and enables the service for the currently logged-in user only.

```
systemctl --user enable --now gnome-remote-desktop.service
```

4. Configure the Firewall

Allow incoming RDP connections through the firewall.

```
sudo ufw allow 3390/tcp
```

5. Set RDP Credentials for the User Service

This command does not use sudo or the --system flag, as it configures the service for your user only. Replace "your_password" with a strong password.

```
grdctl rdp set-credentials "your_username" "your_password"
```

6. Restart the User Service

To ensure the new credentials are applied immediately, restart the user service.

```
systemctl --user restart gnome-remote-desktop.service
```

Your machine is now ready to accept remote connections to the currently active session.